

1. 波函数

1. 光电效应 $\Rightarrow$ 粒子性

2. 杨氏干涉实验 $\Rightarrow$ 验证波动性的实验

3. 黑体辐射 $\Rightarrow$ 证实光的量子性

4. 康普顿散射 $\Rightarrow$ 证明光子有动量

5. 戴维孙-革末实验 $\Rightarrow$ 证实电子具有波动性

二. 波函数

三. 波函数

四.

(1) 在 $\hat{H}_0$ 系中:

$$\hat{H}_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \hbar\omega & 0 \\ 0 & 0 & 2\hbar\omega \end{pmatrix}$$

波函数

$$|\psi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, |\psi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |\psi_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

对波函数 $|\psi\rangle$ 有:

$$|\psi\rangle = C_1|\psi_1\rangle + C_2|\psi_2\rangle + C_3|\psi_3\rangle$$

$$= \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix}$$

(2)

(1) 在 $\hat{H}_0$ 系中:

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, |\uparrow\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, |\downarrow\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, |\uparrow\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, |\downarrow\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$|\uparrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$P_{S_x=\frac{\hbar}{2}} = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right|^2$$

$$= \frac{1}{2}$$

$$P_{S_x=-\frac{\hbar}{2}} = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right|^2$$

$$= \frac{1}{2}$$

(3). 求 $\hat{S}_x$ 的本征值

$$\hat{S}_x = \hat{S}_x \cdot \hat{n}$$

$$= \hat{S}_x \cdot \sin\theta \cos\varphi + \hat{S}_y \cdot \sin\theta \sin\varphi + \hat{S}_z \cdot \cos\theta$$

$$= \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \sin\theta \cos\varphi + \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \sin\theta \sin\varphi$$

$$+ \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cos\theta$$

$$= \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta e^{i\varphi} \\ \sin\theta e^{-i\varphi} & -\cos\theta \end{pmatrix}$$

$$\therefore |\uparrow\rangle = \begin{pmatrix} \cos\frac{\theta}{2} e^{-i\frac{\varphi}{2}} \\ \sin\frac{\theta}{2} e^{i\frac{\varphi}{2}} \end{pmatrix}, |\downarrow\rangle = \begin{pmatrix} \sin\frac{\theta}{2} e^{-i\frac{\varphi}{2}} \\ -\cos\frac{\theta}{2} e^{i\frac{\varphi}{2}} \end{pmatrix}$$

$$\therefore P_{S_x=\frac{\hbar}{2}} = |\langle\uparrow|\psi\rangle|^2$$

$$= \sin^2\frac{\theta}{2}$$

(4).

$$\hat{S}_0 = \frac{\hbar}{2} \sin^2\frac{\theta}{2} - \frac{\hbar}{2} \cos^2\frac{\theta}{2}$$

$$= -\frac{\hbar}{2} \cos\theta$$

$$\hat{L} = \frac{\hbar^2}{2I}$$

(1). $\hat{L}^2, \hat{L}_z$ 均与 $\hat{H}$ 对易

故取 $|\hat{L}^2, \hat{L}_z\rangle$ 为 CSB. Q: 方向 $\hat{L} = \frac{\hbar^2}{2I}$ 与 $\hat{H}$ 对易.

$$4\hbar m \cdot \frac{im\varphi}{2\hbar} = m\omega z \cdot r \cdot r \dots$$



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∴ 其波函数为 $\psi(l, m) = Y_{lm}(\theta, \varphi)$
 $l = 0, 1, 2, 3, \dots, E_l = \frac{l(l+1)\hbar^2}{2I}$
 $m \leq |l|$

又 m 简并, 简并度为 $2l+1$

(2). $\hat{H} = A \hat{L}_z \psi$

$\hat{L}_z \psi(l, m) = \hbar m \psi(l, m) + \hbar m \psi(l-1, m)$

基态: $E_0^{(0)} = 0$

$E_0^{(1)} = \langle 00 | \hat{H} | 00 \rangle = 0$

由基态为非简并, 故波函数即为 $|00\rangle$.

第一激发态: $(= \text{简并})$
 $E_1^{(0)} = \frac{\hbar^2}{2I}$

∴ 设 $\psi^{(0)} = C_1 |11\rangle + C_2 |10\rangle + C_3 |1-1\rangle$

$$\begin{pmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix}$$

又: $H_{lm, l'm'} = \langle l m | \hat{H} | l' m' \rangle$
 $= \langle l m | A \hat{L}_z \psi | l' m' \rangle$
 设 $|11\rangle = |11\rangle$
 $|10\rangle = |10\rangle$
 $|1-1\rangle = |1-1\rangle$

$$\begin{pmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix}$$

∴ $H_{11} = \langle 11 | \hat{H} | 11 \rangle = 0, H_{12} = 0, H_{13} = 0$

$H_{12} = \langle 11 | \hat{H} | 10 \rangle = \int_0^{2\pi} A \psi d\varphi = 0$

$H_{13} = \langle 11 | \hat{H} | 1-1 \rangle = \frac{3\pi A}{8}$

$H_{23} = \langle 10 | \hat{H} | 1 \rangle = \int_0^{2\pi} A \psi \cdot e^{-i\varphi} d\varphi = 0$

又 $H_{33} = \frac{3\pi A}{8}$

∴ 基态

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ B & 0 & 0 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix}$$

$E_1^{(0)} = -B, E_2^{(0)} = 0, E_3^{(0)} = +B$

对第一激发态: $|\psi_1^{(0)}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$|\psi_2^{(0)}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

$|\psi_3^{(0)}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$\psi_1 = (1 + \frac{1}{\sqrt{2}}) |11\rangle - \frac{1}{\sqrt{2}} |1-1\rangle, E_1 = \frac{\hbar^2}{2I} - \frac{3\pi A}{8}$

$\psi_2 = |11\rangle + |10\rangle, E_2 = \frac{\hbar^2}{2I}$

$\psi_3 = (1 + \frac{1}{\sqrt{2}}) |11\rangle + \frac{1}{\sqrt{2}} |1-1\rangle, E_3 = \frac{\hbar^2}{2I} - \frac{3\pi A}{8}$

七

设 $|1\rangle = \psi_1, |2\rangle = \psi_2, |3\rangle = \psi_3$

$\langle 1 | \hat{H} | 1 \rangle = \langle 1 | \hat{H} | 2 \rangle = \langle 1 | \hat{H} | 3 \rangle = \hbar \omega_1$

∴ $\langle 1 | \hat{H} | 1 \rangle = \langle 2 | \hat{H} | 2 \rangle = \langle 3 | \hat{H} | 3 \rangle = 0$

且 $\langle 1 | \hat{H} | 2 \rangle = \langle 2 | \hat{H} | 1 \rangle = \langle 3 | \hat{H} | 1 \rangle = \hbar \omega_2$

在基态

$$\hat{H} = \begin{pmatrix} 0 & \hbar \omega_1 & \hbar \omega_1 \\ \hbar \omega_1 & 0 & \hbar \omega_1 \\ \hbar \omega_1 & \hbar \omega_1 & 0 \end{pmatrix}$$

$= \hbar \omega_1 \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

求本征值及本征函数:

$E_1 = -1$

$E_2 = -1$

$E_3 = 2$

$|\psi_1(0)\rangle = \psi_e$

$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle$

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